

Learning When to Change: Symbolic Supervision for Selective Evidence Sensitivity in Medical Reward Models

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Abstract

Reward models decide which medical reasoning traces are kept, reranked and reinforced. A useful reward reverses its preference when decisive patient evidence changes and holds it when similar-looking evidence does not apply: a penicillin allergy in the patient is a reason to change an order, the same allergy in the patient’s mother is not. We test this *selective sensitivity* with symbolic triplets from executable, stated clinical rules: a base case, a flip of one decisive input, and a near-miss that mentions the same concept without making it applicable. Labels state compliance with the rule shown, not clinical optimality. Across 13 reward signals, 14–75% of triplets are solved: trained medical PRMs seldom reverse, and general judges reverse on near-misses. Training on flip pairs alone leaves this over-triggering unresolved, and near-misses in the same budget remove most of it. A reader that records each finding with its subject, status and time, followed by a judge with no direct access to the case, solves 99% of triplets on rules of held-out structure, 1.2 points more than an evidence summary in prose under the same two-stage interface. Trained on rules alone, with no medical QA data, the model raises agreement with MedEinst pair labels from 18% to 45% and macro-F1 against physician-annotated eligibility judgments from *tbd* to *tbd*. Outside rules, selectivity is tested only indirectly, since existing clinical benchmarks contain no case edits of this kind.

1 Introduction

A reward model for medical reasoning is asked whether a step or an answer is right for the patient at hand. Often the answer turns on one line of the case. Under a rule that orders amoxicillin for streptococcal pharyngitis unless the patient is allergic to penicillin, the order is right for a patient with no drug allergy and wrong for one with a documented penicillin allergy (Figure 1). It is right again for a

Stated rule. Strep. pharyngitis: order amoxicillin; if the patient is allergic to penicillin, order azithromycin.

Claims. s : “Order amoxicillin.” s' : “Order azithromycin.”

Case	Reports	Rule	Required d
base x	allergies: none	s	> 0
decisive edit x^f	allergy: penicillin	s'	< 0
near-miss x^n	mother: penicillin all.	s	> 0
missing x^m	allergy status unknown	–	reject both

Figure 1: One rule, four generated cases (schematic). $d = u(s) - u(s')$ is the preference a verifier should show. A verifier that prefers s throughout fails to reverse; one that prefers s' whenever the allergy is named reverses on the near-miss.

patient whose mother has that allergy. A reward that prefers the same order in all three cases fails to reverse when required. One that changes its preference in the second and the third reverses under an edit that leaves the decision unchanged. Whether evidence applies also depends on the rule: a stroke ten years ago counts under a score that asks for any prior stroke and not under a criterion that asks about the last six months.

Both failures matter because reward models filter training traces (Chen et al., 2024), rerank candidates (Snell et al., 2025; Yun et al., 2025) and supply the reward in reinforcement learning (Shao et al., 2024; Guo et al., 2025). The first is documented: LLM judges are more reliably invariant to irrelevant edits than sensitive to relevant ones (Chen et al., 2026b), and clinical LLMs mention a changed patient fact in 85.6% of their rationales while keeping the original medication choice in 69.8% (Hou et al., 2026). The second is easy to overlook: an output change after a targeted edit means little unless it exceeds the change after an edit that should not matter (Yang et al., 2026). Medical reward models are evaluated on answer selection and on injected step errors (Yun et al., 2025; Wu et al., 2026), neither of which changes the case. Training for sensitivity to relevant changes and in-

variance to irrelevant ones exists for reward models (Srivastava et al., 2026), and labels from formal verification transfer to informal tasks (Kamoi et al., 2026). What is not established is what a reward learns when the “irrelevant” evidence is about the very concept the rule tests, and which supervision teaches it when that evidence applies.

We build the test from executable rules (§3). Each rule yields *triplets*: a base case, a *decisive edit* (flip) of one input, and a *near-miss* in which the decisive concept appears without counting under the rule: in a relative, negated, at a time the rule does not count, or as a value that approaches a threshold without crossing it. Each case is crossed with the two claims the rule can support, and the rule program labels all six instances. Labels state what a stated rule implies, not what is clinically best, and each judgment concerns one condition of one rule: the task is local applicability verification. External benchmarks with existing labels form further tiers, each measuring a different construct (Table 1).

Three things should be kept apart. A *claim verifier* scores a claim against a case; a *process reward model* uses such scores to rank or filter reasoning traces; a *training reward* is optimised against by a policy. Our triplet experiments concern the first, candidate selection (§7.5) the second, and policy training (Appendix G) the third. A result at one level is not evidence for the next, and we make no claim about preventing reward exploitation beyond what the policy-training experiment measures. Our contributions:

(1) **A test.** Triplets separate two failures that pair tests conflate, and rule-side edits change whether the same evidence counts. Across 13 reward signals 13.8–75.0% of triplets are solved (§7.1).

(2) **A controlled comparison** of training distributions and intermediate representations under one interface and budget (§7.2): flip pairs alone leave over-triggering unresolved, near-misses correct it, and a typed applicability ledger is 1.2 points above a prose summary that carries the same facts and is read by the same case-blind judge. A separately written record is what helps: the same ledger written in one pass with the verdict gives no gain.

(3) **Transfer under named controls.** A model trained on rule triplets only reaches 99.2% on rules of held-out structure and, without medical QA data, raises agreement with MedEinst labels from 18.4% to 44.8% and with physician eligibility judgments from tbd to tbd (§7.3).

2 Related Work

Validity of evaluators. Chen et al. (2026b) define a judge’s validity as invariance to construct-preserving edits together with sensitivity to construct-changing ones. Yang et al. (2026) show that the effect of a targeted counterfactual edit must be compared with that of meaning-preserving edits of the same size; our near-miss and presentation edits are such baselines with known labels. Robustness work on reward models targets response-side artifacts (Liu et al., 2025; Zang et al., 2025; Xu et al., 2025). CROME trains reward models with augmentations that change causal attributes of a response and with neutral augmentations that should leave the reward unchanged (Srivastava et al., 2026): sensitivity to relevant changes together with invariance to irrelevant ones is established for reward training. DynaCF uses perturbation sensitivity differently: it rewrites the chosen response without changing its content and down-weights training pairs whose margin shifts (Liu et al., 2026); we include an adaptation of this reweighting to case edits as an alternative to training on near-misses. CSR regularises a solver to change its answer when an operator in its own trace is swapped (Shihab et al., 2025).

Process rewards and symbolic supervision. PRMs are trained on human step labels (Lightman et al., 2024), rollout estimates (Wang et al., 2024) or generated verification reasoning (Khalifa et al., 2025), which GenPRM extends with executed code (Zhao et al., 2025). FoVer trains step verifiers on labels from formal tools and shows transfer to informal tasks including NLI (Kamoi et al., 2026): generating supervision by formal verification, and its transfer, are established and are not our claim. We ask which interventions teach applicability and compare with a model trained on FoVer’s data. Other work perturbs steps and keeps a verifier at inference (Zi et al., 2026), injects errors into verified chains (Chi and Wang, 2026), or uses a deterministic verifier as the reward (Pronesti et al., 2026). In medicine, Med-PRM verifies steps against retrieved guidelines (Yun et al., 2025), MedCEG rewards a policy for covering a reference evidence graph built for each training question (Mu et al., 2025), and MedPRMBench injects 14 error types into chains from seven QA sources (Wu et al., 2026). MedGuideX compiles guideline recommendations into executable functions and post-trains a clinical policy on factual and counterfactual ques-

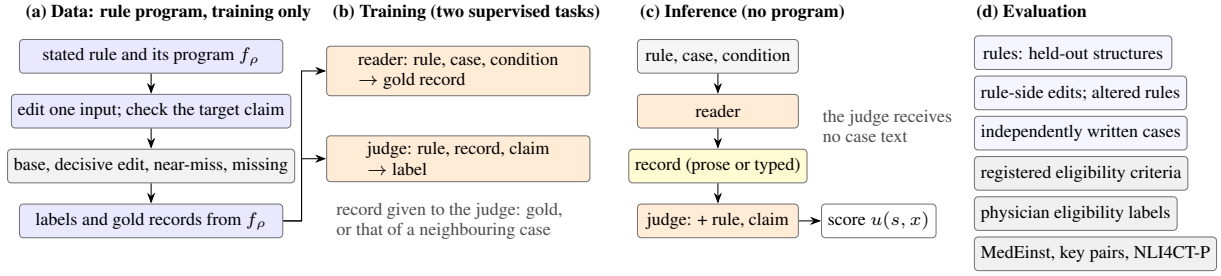


Figure 2: Overview. (a) The rule program produces cases, labels and gold evidence records. (b) It supervises two tasks and is not used afterwards. (c) At inference the reader writes a record from the rule, the case and the condition under test; the judge scores the claim from the rule and the record. A string check on quotations is the only symbolic operation. (d) Controlled tests (blue) and external labels (grey).

tions generated from them, keeping only interventions that change the outcome (Shen et al., 2026); executable guideline functions are also called at inference (Cao et al., 2026). Executable medical logic and counterfactual post-training therefore exist; we compare training with and without the unchanged-outcome cases and do not claim that models trained by that pipeline fail on near-misses, which we have not measured. Mediating a reward through explicit concepts is established (Koh et al., 2020; Laguna et al., 2025) and precedes our ledger, as does EVPV, which extracts visual facts once per problem, independently of the solution, and attenuates the reward of steps whose stated premises they do not support (Wang et al., 2026). The fields of our ledger are the attributes that clinical NLP assigns to mentions of findings: negation, experiencer and temporality (Chapman et al., 2001; Harkema et al., 2009; Uzuner et al., 2011). Typed evidence records and distractors that are similar but irrelevant have been studied in retrieval (Zhou et al., 2026; Agand, 2026; Chen et al., 2026a), conditional QA asks for answers that change with a stated patient condition (Parekh et al., 2026), and criterion-level trial eligibility has physician-annotated judgments (Jin et al., 2024). What we add is a combination: applicability that depends on the stated rule, controlled interventions on decisive and on inapplicable evidence with program-derived labels, and a comparison of supervision and of intermediate representations under one budget.

Context against prior, and counterfactual data.

How language models arbitrate between memory and conflicting context is well studied (Xu et al., 2024; Wu et al., 2024; Shi et al., 2024). Clinical policies keep the prototypical answer when decisive evidence changes (Chen et al., 2026c; Hou et al., 2026; Kim et al., 2025; Qu and Färber, 2026),

and clinical-trial NLI has been probed with label-preserving and label-inverting interventions (Julien et al., 2024). Minimal label-flipping edits serve training (Kaushik et al., 2020) and evaluation (Gardner et al., 2020; Ribeiro et al., 2020; Thrush et al., 2022), with the caveat that narrow edits may not help out of distribution (Joshi and He, 2022); claim-only scorers are partial-input baselines (Poliak et al., 2018).

3 Rule Triplets and External Tiers

3.1 Rule tier

Rules, states, claims. A rule ρ has criteria $c_1, \dots, c_m$ over typed inputs. A generated state z is a set of *mentions*; each has a concept, a value, a subject (patient or other), a status (present, absent, unknown) and a time (current or past). Every criterion c_j comes with an *applicability predicate* A_j that says which mentions it counts. The predicate is part of the rule: a creatinine threshold counts the patient’s current value, a stroke-risk score counts a stroke at any time, and some risk scores count disease in a first-degree relative. In 23% of criteria a past or family finding counts, so no global heuristic about time or subject is correct. When a concept is mentioned more than once, a criterion uses the mentions its predicate counts: the most recent current value of a measurement, any counted mention of a finding with status present. Units, threshold inclusivity, the reference date and window boundaries are fixed per rule and stated in its text (Appendix C). *Scoring rules* are clinical calculators, with claims of the form “criterion j adds b points”. *Constraint rules* state a decision in full: choose a unless condition c holds, then b ; the claims are the two options in neutral wording of matched length. The prompt states the rule. A contraindication can exclude an order, and no single

rule establishes that an alternative is optimal; our labels claim only the former kind of fact.

Edits and invariants. The engine edits the mentions of one input of a criterion c_j and keeps an edit only if executing the program confirms its invariant. A *flip* (decisive edit) changes a mention that A_j counts so that c_j changes, no other criterion does, and the target claim changes; the engine checks the claim itself, since in a rule that combines criteria a changed criterion need not change the decision. A *near-miss* leaves every criterion and the claim unchanged; it is *numeric* (a counted value moves toward the threshold without crossing; a value exactly at the threshold is reported as its own kind) or concerns *applicability* (the concept and value of the flip are reported in a mention that A_j does not count: another subject, absent status, or a time outside A_j). A *presentation* edit re-renders the same state with another unit, order or template, matched in size to the flips (Yang et al., 2026). For *missing* twins we distinguish explicit absence, which determines a criterion, from an unknown input. With $C_\rho(z')$ the non-empty set of admissible completions of a partial state z' , a claim s is *supported* if every completion supports it, *refuted* if every completion refutes it, and *undetermined* otherwise; inconsistent states are excluded. A missing twin is kept only if the target claim is undetermined, which an unknown criterion does not by itself guarantee. A verdict “+” means supported; “−” covers refuted and undetermined claims, which verifiers for retrieval keep apart (Qiu et al., 2026; Kamelhar, 2026) and we report separately. How omitted information is read is a stipulated convention of the generator (measurements unknown, history items absent unless the case says otherwise; Appendix C); external datasets keep their own conventions.

Tests the renderer does not determine. *Rule-side edits* keep the case and change only the applicability clause of the rule (“ever” against “within the last six months”; “the patient” against “the patient or a first-degree relative”), so that the same evidence counts under one rule and not under the other; an item crosses two rules with three cases and is solved only if all six judgments are right (crossed accuracy, XA). *Author-written cases* are written from a state specification without access to generated text. *Rewritten cases* are model rewrites kept only when two extractors recover the state; rejected rewrites are an audit set and are not scored (Appendix C).

Triplets. A triplet crosses a base case x , its flip x^f and a near-miss x^n with the claims s (correct for x and x^n) and s' (correct for x^f). For a reward with score $u(s, x)$ let $d(x) = u(s, x) - u(s', x)$. The triplet is *solved* if $d(x) > 0$, $d(x^f) < 0$ and $d(x^n) > 0$; ties are failures.

Proposition 1. *Let u be deterministic. (i) If $u(s, x) = g(s)$, then $d(x) = d(x^f)$. (ii) If $u(s, x) = g(x)$, then $d \equiv 0$. (iii) If $u(s, x) = g(s, h(x))$ for a feature map h with $h(x^f) = h(x^n)$, then $d(x^f) = d(x^n)$. In each case the triplet is not solved; under (iii) a correct reversal on the flip implies failure on the near-miss.*

Proof. (i) and (ii) are immediate from the definition of d . (iii) $d(x^f) = g(s, h(x^f)) - g(s', h(x^f)) = g(s, h(x^n)) - g(s', h(x^n)) = d(x^n)$, while a solved triplet needs $d(x^f) < 0 < d(x^n)$. $\square$

Let $h_k(x) \in \{0, 1\}$ indicate whether the decisive concept k is named anywhere in the case, whatever its value, subject, status or time. For a finding that the flip introduces, the base case is rendered without naming k and an applicability near-miss names it, so $h_k(x) = 0$ and $h_k(x^f) = h_k(x^n) = 1$: (iii) applies. For a measured input k is named in all three cases, so $d(x) = d(x^f)$ as in (i). A scorer that uses the case only through h_k thus solves no triplet, although it can reverse on every flip of the first kind. Richer attribute-blind features need not collide and are an empirical baseline (Appendix F). Flip pairs alone exclude only (i) and (ii). The proposition concerns deterministic scorers and the stated feature collisions; it is a check on the construction and does not exclude other shortcuts. It does not extend to sampled decisions: three independent fair preference signs match the required pattern with probability 1/8, so sampled readouts are reported with tie rates and intervals.

Metrics. *Reversal* (Rev): share of base–flip pairs with $d(x) > 0$ and $d(x^f) < 0$. *Hold*: share of near-misses with $d(x^n) > 0$, that is, near-miss correctness; it does not by itself show that a correct base judgment was preserved. *Triplet accuracy* (TA): share of solved triplets, the joint endpoint and primary summary. None needs a threshold. Appendix F separates unchanged decisions, jointly correct decisions and preservation given a correct base, and gives base accuracy, ties and denominators per near-miss kind. For missing twins, MR is the rate at which both claims are rejected at the threshold that gives 5% false rejection of supported claims on development data.

Source	Label basis	What it can show
Rule triplets	executed program of a stated rule	reversal and holding under that rule
Registered criteria	program of a registered trial eligibility criterion, checked by an author	the same, on authentic rule language
TrialGPT annotations	physician level labels and evidence sentences	agreement with expert eligibility judgments
MedEinst	DDXPlus cases, edited and filtered by the benchmark; a sample reviewed by physicians	agreement with labels across paired diagnostic cases
Key pairs	answer keys of two exam questions	paired answer discrimination
NLI4CT-P	expert labels, controlled interventions	response to statement edits, report fixed

Table 1: Sources and constructs. Only the first two contain controlled edits; the others are external labels.

3.2 External tiers

Program-derived labels establish that a claim complies with the stated rule. Whether the text expresses the state, whether the test measures applicability and not artefacts of the generator, and whether the behaviour matters outside the generator are separate questions (Appendix B). For the last, labels that we generate with one model and approve with another would not be ground truth, so the external tiers use only released labels. Each source supports a different statement (Table 1).

Registered criteria are unambiguous registered clinical-trial eligibility criteria quoted as written, with programs checked by an author; a selected subset, from which nothing follows about trial eligibility in general. *TrialGPT annotations* (Jin et al., 2024) give physician judgments for 1,015 patient-criterion pairs from 53 patient summaries; we use the expert labels only as targets, keep the dataset’s categories, and resample by patient (protocol in Appendix D). It is an external expert-annotated benchmark, not clinical validation: a gain shows better agreement with expert criterion judgments, not better diagnosis, prescribing or trial screening, and we prioritise it over MedEinst because its task, a stated criterion applied to a case, matches ours. We commission no new clinical annotation: generated examples are validated against explicit specifications, and transfer is evaluated on existing expert annotations. *MedEinst* (Chen et al., 2026c) pairs a control case with a trap case that differs in discriminating findings and has another diagnosis

(5,383 test pairs; 10,689 reference pairs). *Key pairs* are two exam questions whose keyed answers each occur among the other’s options. A claim is the complete question with one candidate answer, so the four labels follow from the keys. We exclude negatively phrased questions and options that refer to other options, and keep each question in at most one pair. Two exam questions are not a controlled counterfactual; the test measures discrimination between answers given the case. We mine pairs from MedQA (Jin et al., 2021) and CareQA (Arias-Duart et al., 2025). *NLI4CT-P* (Jullien et al., 2024) edits expert-written statements about trial reports; we report its faithfulness and consistency with base F1.

4 Diagnostics

Parts and whole. For each base-flip pair we build a *reading* pair, whose claims quote the decisive finding, and an *application* pair, which scores the original claims against a one-line case with only that finding. With P, K, U the events of a correct reversal on the three pairs, the composition gap is $G = 1 - \Pr[U \mid P \wedge K]$; it is descriptive, since the tasks differ in length and in what must be selected.

Appendix F analyses how far, and in which direction, a decisive edit moves the preference relative to a missing-input baseline.

5 Ledger-RM

Reader. The reader R_ϕ receives the case x , the rule ρ where one is stated, and the *condition under test* ω : the criterion, or the set of candidate answers, on which the claims disagree. It does not receive a candidate’s asserted outcome, so the claims of a triplet are judged from one ledger $e = R_\phi(x, \rho, \omega)$ of up to 3 entries. An entry records facts and no decision: the condition the claim depends on (*need*); what the case says about it (*found*: a value with unit, a verbatim quotation, or *missing*); whose finding it is (*subject*); its *status* (*present*, *absent*, *unknown*); and its *time* (*current* or *past*). Keys and categorical fields are fixed by a grammar, and a quotation that does not occur in the case makes the ledger malformed. On the rule tier ω is supplied: it names the criterion on which the two claims disagree and carries no label and no information about the edit. A deployed verifier would have to find it, so an ablation has the reader derive ω from the rule and the claim (§7.4). Because the reader does not see the asserted outcome, opposing

claims about one condition are judged from the same record.

Judge. The judge J_ϕ receives the ledger, the rule if any, and the claim, and decides whether the recorded finding counts under the rule and what follows for the claim. The end-to-end score is

$$u(s, x) = J_\phi(R_\phi(x, \rho, \omega), \rho, s), \quad (1)$$

with $J_\phi(e, \rho, s) = \log p_\phi(+ | e, \rho, s) - \log p_\phi(- | e, \rho, s)$. The judge has no direct case input: the case reaches the verdict only through the ledger. This restricts the channel and does not guarantee a faithful judgment, since a reader could encode cues in field values, their order or the quotation; field interventions and paraphrase controls test this (§7.4). A malformed ledger is an explicit invalid outcome: both claims are rejected and the pair is a tie, which counts as a failure (log-odds are unbounded, so no finite floor is used). Reader and judge are two prompts of one adapter.

Training. On rules the program supplies the target ledger $e^*(x, \rho, \omega)$ and, for any ledger e , the verdict $y_\rho(e, s)$ obtained by executing the rule on the state that e describes. Let $N(x)$ be the other training cases of the same rule and condition, including the twins of x . The judge is trained on a ledger $e \sim q(\cdot | x, \rho, \omega)$, which equals $e^*(x, \rho, \omega)$ with probability $1 - p$ and $e^*(x', \rho, \omega)$ for x' drawn uniformly from $N(x)$ otherwise:

$$\mathcal{L} = \mathbb{E}_{(x, \rho, s)} \left[-\frac{1}{|e^*|} \log p_\phi(e^* | x, \rho, \omega) - \lambda \mathbb{E}_{e \sim q} \log p_\phi(y_\rho(e, s) | e, \rho, s) \right]. \quad (2)$$

A resampled ledger is a valid input for the judge, whose target is the verdict that ledger implies; no instance rewards a verdict that contradicts the evidence shown to the judge. In implementation the two terms are separate supervised examples in one corpus: one reader example per case and condition, shared by both claims, with $|e^*|$ the number of target tokens, and one judge example per claim; λ is their ratio in the corpus. A deterministic parser maps ledger text to a state for y_ρ ; incomplete or inconsistent ledgers have no state and yield no judge example. Donor ledgers are gold ledgers, whereas at test time the judge reads generated ones; the program-supplied-ledger analysis measures that gap.

What follows by construction, and what does not. The verdict is computed from the ledger. That the ledger is correct and sufficient, and that the judge uses its fields as intended, is learned; we test the latter by field interventions and by ablations that add or isolate a decision field (§7.4). A missing or misattributed entry cannot be repaired by the judge. For the same items we report the paired transitions between the model’s own ledger and a program-supplied one (errors rescued, correct judgments broken, errors unchanged); the gain bounds what better extraction could recover and does not by itself assign causal responsibility. We also apply the rule program to the predicted ledger, which shows what the learned judge adds when a program is available, and test a second pass that re-reads every entry against the case (Agand, 2026).

Step check and combined reward. Errors internal to a step are scored by a separate adapter, the injected-error PRM, which sees the case. A trace is acceptable only if both checks pass. For candidate selection we rank by the minimum of the two scores after temperature scaling on development data. The minimum is a heuristic: it is an upper bound on the joint probability, not that probability, and scaling each score separately does not account for dependence between the two checks. We evaluate it against the product, a combination fitted on development data and reliability gating (Wang et al., 2026) (Appendix G). In selection a step is scored alone, so the reader receives the step and the ledger can depend on it (Appendix E). All other results use the ledger score alone, and the ledger adapter is never trained on step-error or QA data unless a row says so.

6 Experimental Setup

Rules and shifts. The frozen library (rule_v1) has 353 rules: 43 scoring rules that we implemented from published definitions, 60 constraint rules over renal function, pregnancy, allergy, age and interacting drugs, compiled from public recommendations by the tree-then-program procedure of Shen et al. (2026), and 250 rules sampled from a typed grammar. MedCalc-Bench calculators (Khandekar et al., 2024) are not included in this version. A manifest records for every rule its source or specification, the text shown in the prompt, the program, its applicability semantics, its tests and its version, and whether the text is a verbatim source rule, a simplified source rule or synthetic (Ap-

pendix B). Every program passes boundary, unit and dependency tests against the rule text. Each rule has a *structural signature* (rule kind, operators, dependency depth, input types, applicability predicate). The test levels separate four kinds of novelty: new generated cases of trained rules (L0); new rule identities and constants within seen signatures (L1); held-out signature classes, that is, unseen logical compositions (L2); and trained rules whose thresholds are changed in the prompt, restricted to cases on which the original and the altered rule disagree (L3-alt). In all test sets the rendering templates and cue phrases for relatives, negation and time do not occur in training, and all cases derived from one base state stay in one split. Rule-side items, author-written cases, rewritten cases and registered criteria (§3) are separate, separately frozen sets. Structure statistics and overlap at the level of program, normalised structure, subexpression, template and cue phrase are in Appendix B.

Training distributions. All trained models use Qwen3.5-9B with LoRA (Hu et al., 2022) of rank 64 for one epoch on 60,000 records. Four corpora share one record format and the same share of missing-input and presentation records (15% each) and differ in which cases they contain. *Natural*: one case per generated group, drawn so that the decisive condition is rare; the rate is a parameter of the generator, not an empirical prevalence. It is a simplified stand-in for step-perturbation corpora (Zi et al., 2026; Chi and Wang, 2026), which we do not reimplement. *Balanced*: one case per group, with the two outcomes equally frequent. *Blocks*: the base case and the decisive edit of every group. *Triplets*: blocks in which a near-miss replaces the base case in half of the groups. The loss is computed per example, so the corpora differ in their cases and not in the objective or the batching; pairing as such is never observed by the optimiser. Appendix G gives, per configuration, unique cases, claims, reader and judge targets, tokens, optimiser steps and runtime.

Representations. Two one-stage formats, in which the verdict is produced with the case in context: verdict only, and a rationale (the ledger) followed by the verdict. Three two-stage formats with the same case-blind judge interface: an evidence summary in prose, a value ledger (need, found), and the applicability ledger. The prose summary is rendered from the same supervised facts as the applicability ledger, including subject, status and

time, so that the two differ in form and not in information; neither contains a verdict, which we verify on generated records. A sixth variant gives the judge of the prose pipeline the case as well. Table 16 (appendix) states what each comparison isolates. Budgets are matched in records and reported in actual tokens at training and inference.

Systems. *Audited signals*: Med-PRM (Yun et al., 2025), the MedS³ PRM (Jiang et al., 2025), the FoVer PRM (Kamoi et al., 2026), a PRM trained on the MedPRMBench training split (injected-error PRM), ThinkPRM (Khalifa et al., 2025) and GenPRM-7B with code execution (Zhao et al., 2025); Qwen3.5-9B, Qwen3.5-27B, Llama-3.3-70B and Kimi K3; GPT-5.4, Gemini 3.1 Pro and Claude Opus 5.5. *Trained comparisons* on Qwen3.5-9B: FoVer’s training data; the five representations; and a GenPRM-style verifier, trained on the same triplets to write an analysis and a Python check of the claim, whose executed output precedes the verdict (targets rendered from the rule program). MedCEG’s reward needs a reference graph for the question, so it cannot score a claim alone; we use one of that kind only in policy training (Appendix G). *Training-free*: default correction, $u(s, x) + \alpha [u(s, x) - u(s, \emptyset)]$ (Shi et al., 2024), which amplifies the contribution of the case; a prompted ledger; and a verifier that writes and executes a program for the stated rule (Ocampo Osorio et al., 2026). *References*: the best closed judge with the prompted ledger; extraction followed by a hand-written program, which uses the rule program at test time, a resource no other system has.

External data and protocol. Registered eligibility criteria and the TrialGPT criterion annotations (§3); MedEinst test pairs, 1,240 key pairs from MedQA test and 960 from CareQA, and NLI4CTP; versions are pinned (Appendix D). Scores are verdict log-odds where token log-probabilities are available and verdict frequencies over 32 samples otherwise. We train 5 seeds; values printed in black in this draft are from seed 0. Systems are compared on identical items with paired differences; resampling keeps triplets together (patients for TrialGPT) and treats rules as clusters. We list every seed and keep apart the uncertainty from the evaluation sample and the variation across seeds, report macro-averages over rules and signature classes, and give denominators for each near-miss kind. Overlapping intervals are not read as a test of difference or of equivalence, and near the ceiling we

Reward signal	Rule triplets				ME Key	
	Rev	Hold	TA	MR	Rev	Rev
<i>Trained PRMs</i>						
MedS ³ PRM	14.9	91.0	13.8	9.5	12.1	33.0
Med-PRM	19.6	89.5	17.7	12.8	15.3	40.2
FoVer PRM	22.4	87.2	19.8	16.0	11.0	24.7
Inj.-error PRM	23.8	90.1	21.5	15.2	13.6	36.9
ThinkPRM	30.2	71.5	22.4	20.3	14.8	38.5
GenPRM-7B	33.5	73.0	25.6	22.0	15.5	39.0
<i>Open judges</i>						
Qwen3.5-9B	37.5	66.3	27.9	24.1	18.4	44.5
Llama-3.3-70B	58.9	68.0	42.1	41.5	33.0	60.7
Qwen3.5-27B	62.0	70.2	45.3	44.9	35.8	63.2
Kimi K3	68.7	72.4	51.0	50.6	41.2	68.0
<i>Closed judges</i>						
GPT-5.4	84.1	83.0	71.0	66.2	55.6	79.9
Gemini 3.1 Pro	85.0	84.1	72.6	67.0	57.1	81.0
Claude Opus 5.5	86.3	85.9	75.0	68.0	58.3	82.4

Table 2: Audit (%), each signal with its native readout (Appendix F). Rule triplets, L0: reversal, hold, triplet accuracy, missing-evidence rejection at 5% false rejection. ME, Key: reversal on MedEinst and on key pairs.

state absolute differences. Because the held-out signature classes are few, thousands of generated cases are not thousands of independent observations about structural generalisation. `rule_v1` was frozen before the runs reported here and is not modified; sets built later are separate named versions with their own frozen tests (Appendix B). Primary endpoints are triplet accuracy on L2 and MedEinst reversal of the rule-only model, with 5 Holm-corrected comparisons; systems and thresholds are selected on development splits and frozen before test (Appendix A).

7 Results

7.1 Audit of existing reward signals

The two failures separate the signals (Table 2). Trained PRMs seldom reverse (14.9–33.5%) and therefore hold; they solve 13.8–25.6% of triplets. Open judges reverse more often and hold on only 66–72% of near-misses. Closed judges solve 71.0–75.0%. Reversal is lower on MedEinst for every signal and higher on key pairs, which are not minimal. Shortcut scorers show what the tested shortcut implementations can reach, not that no shortcut exists (Appendix F): a classifier that sees only the step reaches a macro-F1 of 81.7 on the MedPRMBench test split, against 41.2 for the majority class, and solves no triplet; a scorer that sees only whether the decisive concept is named solves no triplet on L2; a bag-of-words model solves 7.1%; a logistic model on attribute-blind concept–value features

Representation	Nat.	Bal.	Blocks	Triplets
<i>One-stage (verdict produced with the case in context)</i>				
Verdict only	58.0	49.1	58.2	91.3
Rationale, then verdict	55.1	tbd	61.5	91.0
<i>Two-stage (case-blind judge)</i>				
Evidence summary (prose)	tbd	tbd	tbd	98.0
Value ledger	tbd	tbd	tbd	tbd
Applicability ledger	tbd	tbd	75.3	99.2
<i>Two-stage, judge also sees the case</i>				
Evidence summary	–	–	–	tbd

Table 3: Triplet accuracy on L2 (%) by training corpus and representation. Qwen3.5-9B, rule data only, 60k records, one epoch. Values in black are measured (seed 0); intervals over rules: verdict 58.0 [54.1, 61.9], 49.1 [45.6, 53.1], 58.2 [54.2, 62.5], 91.3 [87.3, 95.0]; summary 98.0 [97.0, 98.9]; ledger 75.3 [72.0, 78.5], 99.2 [98.6, 99.7]. They reflect variation between rules, not between seeds. Corpora differ in which cases they contain (§6).

solves 40.0%, all of them numeric triplets; and a trigger lexicon built from training cue phrases solves 54.0%. Given the test cue phrases the same lexicon solves 99.5%: the difficulty of the generated test lies in unseen wording and rule structure, which is why the tests of §3 that the renderer does not determine matter.

Every signal reverses correctly on at least 84.9% of reading and of application pairs and fails the full pair in 11–80% of the cases in which it solves both (Appendix F).

7.2 Training distribution and representation

Flip pairs alone leave over-triggering unresolved. At seed 0, trained on blocks, the verdict-only model reverses on 93.1% of flips and is right on 61.5% of near-misses; its triplet accuracy (58.2%) is close to that of the trigger lexicon built from training cue phrases (54.0%). Replacing the base case by a near-miss in half of the groups raises near-miss correctness to 97.7% at a reversal of 92.3% and triplet accuracy to 91.3% (Table 3). The one-stage rationale moves from 61.5% to 91.0% and the applicability ledger from 75.3% to 99.2%. The statement is limited to these three formats: the matched prose pipeline has no blocks run yet (tbd). Without near-misses the failures are concentrated where the finding belongs to another person (11.5% of such triplets solved by the verdict-only model) or the value sits on the threshold (33.8%); negated findings and values near the threshold are at 92–93%. Data from a pipeline that keeps only interventions that change the outcome (Shen et al., 2026), or

from an objective that rewards change under an edit (Shihab et al., 2025), contain no unchanged-outcome cases by design; we test what adding such cases changes and have not tested those systems. Using near-misses only as probes for re-weighting, in an adaptation of DynaCF to case edits, gives **tbd** (Table 14).

Does balance teach reversal? We expected balanced outcomes to teach reversal. At seed 0 the balanced corpus is below the natural one (49.1% against 58.0%) and blocks are level with it (58.2%); the remaining seeds and the reversal rates of these two corpora (**tbd**, **tbd**) decide whether that expectation holds. The loss is per example, so blocks differ from the balanced corpus only in containing both members of each pair.

One stage against two. With near-misses, the two formats that judge from a separately written record reach 98.0% (matched prose) and 99.2% (applicability ledger); the rationale, which writes the same ledger before the verdict in one pass, reaches 91.0%, level with verdict only (91.3%). So far, then, it is not writing evidence first that helps. The two-stage formats differ from the rationale in three ways at once: a second pass, a judge that does not see the case, and separately supervised reading. The case-visible judge (**tbd**) and Table 14 separate them. The rationale result concerns this rationale format and training setup.

Applicability structure against a matched two-stage baseline. At seed 0 the ledger is 1.2 points above the prose record that carries the same facts. Both are near the ceiling of this test. The paired difference on identical triplets is **tbd** [**tbd**, **tbd**]; the ledger is right and the prose record wrong on **tbd** triplets, and the reverse on **tbd**. We treat the contribution of typed structure as an open empirical question and look for it under shift (Table 4). Without near-misses the ledger is 17.1 points above verdict only (75.3% against 58.2%), mostly on values at the threshold (72.0 against 33.8) and findings in another person (48.5 against 11.5); with near-misses the difference is 7.9 points. The data therefore do not show that ledger and near-misses reinforce each other, and the ceiling limits what this comparison can show.

7.3 Transfer within and beyond rules

Within rules. Trained on rule triplets only, Ledger-RM solves 99.2% of triplets on rules of held-out structure and 97.9% on invented rules; the verdict-only model trained on the same triplets

solves 91.3% and 96.9%, and 99.7% on development rules, so held-out structure costs it about eight points (Table 4, Table 15). Altered rules are not harder in this version: both triplet-trained models solve all of them and the verdict-only model trained on blocks solves 94.6%, far above its L2 score, so that set does not test what we intended (its composition is in Appendix B). The generated test is close to solved by the best system, and its role is the controlled comparison of §7.2. Whether the behaviour is more than a property of the generator is carried by the three tests the renderer does not determine: crossed accuracy on rule-side items is **tbd** for Ledger-RM and **tbd** for the verdict-only model; on author-written cases triplet accuracy is **tbd** and **tbd**; on verbatim eligibility criteria **tbd** and **tbd**. Training on FoVer’s formal data gives **tbd** on L2, and default correction **tbd**.

From rules to clinical labels. The rule-only model has seen no medical QA data of any kind. Against physician criterion-level eligibility labels its macro-F1 is **tbd**, with **tbd** for the untrained backbone and **tbd** for the model trained on blocks, and its quoted evidence matches the sentences the physicians marked with precision **tbd** and recall **tbd**. A gain there shows better criterion judgments on that benchmark; selective sensitivity under controlled edits is established separately, by the triplet and rule-side tests. Its agreement with MedEinst pair labels rises from 18.4% to 44.8% [39.9, 49.6], against 39.4% for the free-form summary. Adding MedPRMBench step-error data to the adapter adds 3.8 points, which is medical supervision and not transfer. Near-misses buy holding at a small price in reversal: against the blocks model, the triplets model loses 1.5 points on MedEinst and 1.4 of NLI4CT-P faithfulness and gains 6.6 of consistency. Trained on clinical pairs, the model reaches 66.8% on MedEinst; this is in-distribution, and with diseases held out it is 58.9% (Appendix D).

Alternatives. A GenPRM-style verifier trained on the same triplets writes and executes a check for every claim. It reaches **tbd**% on L2 and **tbd**% on altered rules, at 11 times the generated tokens; on MedEinst, where there is nothing to execute, it is at 35.8%. Executed code settles how a rule is applied and not how the case is read: in 81% of its near-miss errors a relative’s or a past finding is passed to the code as the patient’s. The closed judge with a ledger prompt leads on L2, key pairs and NLI4CT-P.

System	Rule tier					External		
	L2	L3-alt	XA	Hold	MR	Criteria	TrialGPT	MedEinst
<i>Training-free, Qwen3.5-9B</i>								
Critic	tbd	tbd	tbd	tbd	tbd	tbd	tbd	18.4
+ default correction	tbd	tbd	tbd	tbd	tbd	tbd	tbd	27.0
+ prompted ledger	tbd	tbd	tbd	tbd	tbd	tbd	tbd	26.1
Generated-program verifier	tbd	tbd	tbd	tbd	tbd	—	—	—
<i>Trained without medical QA data (clinical columns are zero-shot)</i>								
Verdict only, rule blocks	58.2	94.6	tbd	61.5	≈100	tbd	tbd	tbd
Verdict only, FoVer data	tbd	tbd	tbd	tbd	tbd	tbd	tbd	19.9
Verdict only, rule triplets	91.3	100.0	tbd	97.7	≈100	tbd	tbd	33.0
Evidence summary, rule triplets	98.0	tbd	tbd	tbd	≈100	tbd	tbd	39.4
GenPRM-style verifier, rule triplets	tbd	tbd	tbd	tbd	tbd	tbd	tbd	35.8
Ledger-RM, rule balanced	tbd	tbd	tbd	tbd	tbd	tbd	tbd	37.2
Ledger-RM, rule blocks	75.3	85.3	tbd	76.2	≈100	tbd	tbd	46.3
Ledger-RM, rule triplets	99.2	100.0	tbd	100.0	≈100	tbd	tbd	44.8
<i>With medical data</i>								
Ledger-RM, rule triplets + step-error data	tbd	tbd	tbd	tbd	tbd	tbd	tbd	48.6
Ledger-RM, clinical pairs only	tbd	tbd	tbd	tbd	tbd	tbd	tbd	62.4
Ledger-RM, rule triplets + clinical pairs	tbd	tbd	tbd	tbd	tbd	tbd	tbd	66.8
<i>References</i>								
Claude Opus 5.5, zero-shot	tbd	tbd	tbd	tbd	tbd	tbd	tbd	58.3
Claude Opus 5.5, prompted ledger	tbd	tbd	tbd	tbd	tbd	tbd	tbd	63.0
Extraction + hand-written program	tbd	tbd	tbd	tbd	tbd	—	—	—

Table 4: Transfer (%), ledger score alone. L2, L3-alt: triplet accuracy on rules of held-out structure and on altered rules; XA: crossed accuracy on rule-side items; Hold, MR: on L2; Criteria: triplet accuracy on verbatim eligibility criteria; TrialGPT: macro-F1 against physician criterion-level labels; MedEinst: reversal. Key pairs and NLI4CT-P for the same rows are in Table 10. Black: measured, seed 0; rule-tier cells not yet measured are marked; external columns keep earlier placeholder values for MedEinst. MR is at the ceiling for every trained model and does not separate them. Clinical pairs: MedEinst reference pairs and MedQA-train key pairs, so MedEinst is in-distribution for those rows. The two verifier rows execute code written by the model; the last row executes a hand-written program on an extracted state, uses the rule program at test time, and its gap to 100 is extraction error.

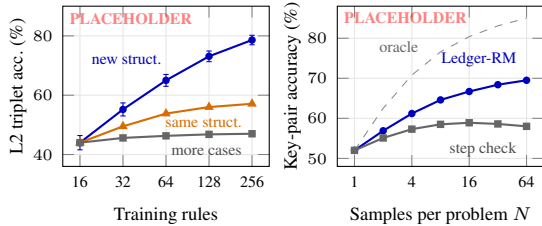


Figure 3: Not yet measured; curves show the layout only. Left: L2 triplet accuracy of Ledger-RM as training grows by adding rules of new structures, rules of structures already seen, or only cases for 16 fixed rules (same data size; bars: s.d. over seeds). Right: accuracy on both members of a MedQA key pair for the trace selected among N samples.

7.4 What the judge uses

Does a decision bit explain the rule-tier result? If the reader also writes whether the condition applies, a judge that sees only that bit and the claim reaches **tbd**% on L2, against 99.2% for the fact-only ledger. If the two are close, the benefit on rules lies in

what the reader is trained to establish and not in the judge’s use of the fields. Transfer is the other test of the fields: **38.2%** against **44.8%** on MedEinst zero-shot (Table 14), and a reader trained to write the bit without fields reaches **tbd**%.

Own against program-supplied ledgers. With program-supplied ledgers the judge solves **tbd**% of L2 triplets. On the same items, **tbd** errors of the full model are rescued by the program-supplied ledger, **tbd** correct judgments are broken and **tbd** errors are unchanged. The first number bounds what better extraction could recover; it does not assign every error to the reader. Applying the rule program to the predicted ledger gives **tbd**%, against 99.2% for the learned judge: if the program matches the judge, what the pipeline contributes on executable rules is learning to extract applicable evidence. Malformed ledgers are at most 0.17% of records. A second pass that re-reads each entry against the case and regenerates rejected ones gives **tbd**%.

One stage against two. The one-stage variant, in

which the ledger is written as a rationale and the verdict is produced with the case in context, solves 91.0% of L2 triplets against 99.2%. It is right on 99.6% of near-misses, so its deficit lies in reversal and not in holding (reversal: **tbd%**). On MedEinst zero-shot it is **2.7** points better than the two-stage model: where a few typed fields cannot carry the evidence, the bottleneck costs accuracy. With the prose record, a judge that also sees the case reaches **tbd%** against 98.0% without it.

Field interventions and the condition under test.

Changing subject, status or time in a correct ledger changes the verdict as the program prescribes in **97.3%** of cases, and given the ledger of another case the judge returns the verdict that ledger implies in **96.8%** (Appendix E). When the reader derives the condition under test from the rule and the claim, triplet accuracy is **tbd%**.

Near-miss kinds not seen in training. Trained on triplets from which one kind of near-miss is removed, the verdict-only model solves **tbd%** of L2 triplets of that kind (another person), **tbd%** (past) and **tbd%** (threshold); the applicability ledger **tbd%**, **tbd%** and **tbd%**. This tests whether near-miss supervision has to enumerate the kinds, and whether a record helps where it was not enumerated.

Rule diversity, model size, missing inputs. Figure 3 (left; not yet measured): rules of new signature classes raise L2 accuracy from **tbd%** to 99.2%, rules of seen classes to **tbd%**, more cases to **tbd%**. With the 4B model of the same family the verdict-only model trained on blocks reaches 49.9%, against 58.2% for the 9B model; the other cells are not run, so we draw no conclusion about scale. Missing-evidence rejection is at the ceiling for every trained model, since the marker of a missing input is learnable from the 15% of such records in each corpus; we report this as a limit of that test and not as evidence of abstention.

7.5 Candidate selection

Table 5 tests the verifier as a process reward: selection among samples of a fixed policy. It says nothing about a policy optimised against the reward (Appendix G). The combined reward and the step check improve accuracy on MedQA and CareQA by similar amounts (difference **0.7** [**-0.2**, **1.6**] on MedQA, non-inferior at a margin of one point). On both members of a key pair the combined reward reaches **66.7%** against **58.9%**, and on MedEinst pairs **45.9%** against **34.0%**, with control accuracy unchanged. Swapping in the vignette

Selection	Accuracy		Key pair	MedEinst		
	MQA	CQA		pair	ctrl	trap
Single sample	71.2	66.8	52.0	31.2	78.5	39.8
Self-consistency	74.5	70.1	55.8	33.5	81.0	41.4
Step check	77.6	72.4	58.9	34.0	81.2	41.9
vignette swapped	76.8	71.9	57.6	33.6	80.9	41.5
Ledger score	76.0	71.0	65.1	44.7	81.3	54.9
Combined	78.3	73.0	66.7	45.9	82.0	55.6
vignette swapped	73.9	69.2	54.1	32.8	80.1	40.6
no near-misses	77.1	72.2	64.9	46.5	81.0	56.8
Claude Opus 5.5	79.5	76.0	70.2	44.1	82.0	53.8
Oracle selection	88.0	85.3	80.4	58.7	90.2	65.1

Table 5: Reranking 16 samples of a Qwen3.5-9B policy on identical pools (%). MQA: MedQA; CQA: CareQA. Step check: injected-error PRM. Ledger score: Ledger-RM trained on rule triplets and clinical pairs. Combined: minimum of the two after calibration.

of another question removes the gain of the combined reward and leaves the step check almost unchanged: the ledger score depends on the supplied case. That the dependence is correct is what the pair columns show. Without near-misses in training the combined reward loses **1.2** points on MedQA and nothing on MedEinst, whose pairs contain no near-misses. As N grows, pair accuracy under the step check stays flat while the oracle rises (Figure 3, right).

8 Conclusion

On executable rules, reward models fail in two directions. The remedy has two parts: training with near-misses, and a record of what the case says, whose it is and when, from which a case-blind judge decides. Trained on rules alone, this carries to rules of unseen structure and, **in part**, to existing clinical labels. Holding on inapplicable evidence in clinical text is not established.

Limitations

What the labels are. Rule-tier labels establish compliance with a stated rule, its applicability predicates and a rendering grammar. They say what the rule implies, not what is clinically optimal, and that a program produced a label does not show that the text expresses the state, that the test is free of generator artefacts, or that the behaviour matters outside the generator; these are addressed separately (Appendix B). We commissioned no new clinical annotation; reused benchmarks carry the review their authors report. External results are agreement with released labels: the TrialGPT anno-

tations are physician judgments on synthetic patient summaries, MedEinst cases derive from a simulator and were edited and filtered by the benchmark, whose authors report physician review of 1,500 generated pairs, key pairs inherit the errors of exam keys and are not counterfactuals, NLI4CT-P edits statements about breast-cancer trial reports. Nothing here supports clinical use.

The rule language. The test is generated from a finite language of applicability conditions (subject, status, time, thresholds). Held-out signature classes reuse primitives and operators seen in training, there are few such classes, and most judgments concern one condition of one rule: the task is local applicability verification, not clinical reasoning. A trigger lexicon that is given the test cue phrases solves the generated test, so its difficulty lies in unseen wording and structure.

Selectivity outside rules is tested only indirectly. Applicability near-misses with exact labels require a rule program. Beyond rules, holding is tested on author-written cases and registered criteria, which still use our programs, and on the claim side: by NLI4CT-P consistency under statement edits and by expert-reviewed distractors in real discharge summaries that attribute a correct fact to the wrong admission (Kim et al., 2026) (Appendix D). No existing benchmark edits the case in this way; assertion annotations of clinical notes (Uzuner et al., 2011) would allow a test of the reader alone.

The bottleneck. The ledger assumes that a claim depends on a few identifiable findings with a subject, status and time. Claims that weigh many soft findings fit it poorly, and the one-stage variant is better on MedEinst. The judge computes its verdict from the ledger; nothing guarantees that the ledger is right, a quotation can exist in the case and still be the wrong evidence, and hiding the case from the judge does not make its judgment faithful. On the rule tier the condition under test is supplied.

Verifier, process reward, training reward. The triplet results concern a claim verifier. Selection tests it as a process reward for one policy. We do not show that a policy trained against it is safer, and we make no claim about preventing reward exploitation beyond the policy-training experiment.

Rule population. Rules come from our implementations of published scores, constraints compiled from public recommendations and a grammar; templates are cleaner than clinical notes. No clinician reviewed the rules.

Development. The generator and the systems were developed together, and the frozen version followed an earlier one whose failure we describe in Appendix H. Choices informed by earlier failures are listed there.

Weak supervision. For rows trained on clinical pairs, ledger targets come from differences between paired cases and from selection with a frozen judge; they are not verified.

Interpretation and scope. The composition gap and shift classes are descriptive. Measured values in this draft come from one seed and one backbone. All data are in English.

Ethical Considerations

The work uses public benchmarks and generated cases; no patient data were collected. Stated rules in our data are test specifications and must not be read as prescribing guidance. A reward model that appears to check patient evidence could be trusted more than it should be; accuracy on these triplets does not establish safety in care, and the models are released for research on post-training only. Because the engine generates cases on which a given reward fails, it could be used to construct inputs that mislead a deployed evaluator; we see this as an argument for auditing rewards before they are used.

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was seen: (6) on the TrialGPT annotations, the rule-only model trained on triplets against the untrained backbone and against the model trained on blocks. **Decision rules.** If (2) fails, near-misses are reported as an evaluation device only. If (3) fails, the paper is a data-design result and the typed ledger is reported as equal to prose. If (4) and (6) fail, the answer outside rules is negative and the medical claim is withdrawn. If the gain in (4) appears only with step-error data, it is reported as a gain from medical supervision.

Status at seed 0. (1) is not supported: balanced is below natural. (2) holds by a wide margin. (3) is 1.2 points near the ceiling and awaits the paired test. (4)–(6) are not run.

Selection and freezing. Hyperparameters, thresholds, the calibration of the combined reward and the choice of headline system use development splits only: development triplets from training-fold signature classes and a held-out part of the MedEinst reference set. Test sets are generated or opened after these choices are frozen and are evaluated once.

B Rule Library and Splits

Source	Rules	Classes
Published scores, our implementation	43	tbd
Constraint rules from public recommendations	60	tbd
Grammar-sampled rules	250	tbd
MedCalc-Bench calculators	not included	
Total (rule_v1)	353	tbd

Table 6: Rule population of the frozen library.

What the labels establish. A program-derived label answers one of four questions (Table 7); the paper treats them separately.

Question	Evidence
Does the program implement the stated rule?	specification review by a second author; boundary, unit and dependency tests
Does the text express the intended state?	rendering audit by the authors (below); extraction check on rewrites
Does the test measure applicability and not artefacts?	shortcut scorers; rule-side items; author-written cases
Does the behaviour carry beyond the generator?	registered criteria; physician labels; MedEinst

Table 7: Four validity questions and the evidence for each.

Manifest and provenance. The released manifest gives, for every rule, its source or specification, the text shown in prompts, the program, its applicability semantics, its tests and its version. Each rule is marked as a *verbatim source rule* (tbd), a *simplified source rule* (tbd) or *synthetic* (tbd). Formalisations of source-derived rules are checked against the source sentence by an author who did not write them, because a generator that checks its own output can repeat its own error; tbd of tbd were corrected or dropped. Requirements that are ambiguous (for example “adequate organ function”) are excluded and not given an invented interpretation. We do not use the calculator set of MedCalc-Eval (Mao et al., 2025), whose definitions were not available to us. Because every prompt states the rule, correctness is relative to the stated text for all sources.

Rendering audit. Before the freeze, 450 rendered groups were reviewed for label errors [state who reviewed them and what they were asked]; none was found in base, flip, near-miss or presentation cases, and missing-input lines that could be read as “absent” were reworded to “unknown”. Presentation edits that changed a fact, sampled alternatives ruled out by the condition, and near-misses that differed from the flip in more than one attribute were found and removed. The authors separately read tbd groups for dropped negation, wrong subject, ambiguous time expressions, conflicting measurements and omitted exceptions, and found tbd; this checks the rendering and not clinical validity.

Constraint rules and claims. Each rule states a default option, a condition and an alternative. It is derived from a public drug-label or guideline recommendation in three steps taken from Shen et al. (2026), whose own functions are not released: a model extracts the recommendation, writes it as a decision tree, and compiles the tree to a program. We keep a rule if the tree is complete, the program passes the boundary tests and agrees with the tree on sampled inputs, and show the verbalised tree in the prompt. No clinician checks the tree against the source, which is why labels are relative to the stated rule. Options are plain orders of one form that differ by at most 2 tokens and contain no word that gives a reason. Every case reports the same fields with raw values, so the presence of a field carries no label information.

Structure and complexity. Rule counts over-state variety, so Table 8 reports structure. Most judgments concern one condition of one rule; where that is so we call the task local applicability verification.

	Train	L2
Distinct program structures (names, constants removed)	tbd	tbd
Signature classes	tbd	tbd
Conditions needed per judgment (mean; max)	tbd	tbd
Dependency depth (mean; max)	tbd	tbd
Evidence mentions required (mean)	tbd	tbd
Cases with competing mentions of the concept (%)	tbd	tbd
Judgments with a temporal operation (%)	tbd	tbd
Judgments with a numeric comparison (%)	tbd	tbd
Judgments with an exception (%)	tbd	tbd

Table 8: Structural statistics of rule_v1.

Splits and overlap. In each of 3 folds 5 of 19 signature classes are held out for L2. The levels change different things: L0 the generated cases only; L1 rule identities and constants; L2 logical compositions; all test sets the wording. Between training and L2 we report overlap of source documents (tbd), canonical programs (0%), parameter-normalised structures (tbd), criterion subexpressions (12.4%), rule texts (0%), rendering templates and cue phrases (0%), and underlying states (tbd): each input type has 6 templates, 4 for training and 2 for test, and the words that mark relatives, negation and time are split in the same way. All cases derived from one base state are in one split. With training templates and cues on the same L2 rules, triplet accuracy of Ledger-RM is tbd, against 99.2% with held-out ones. L3-alt moves one or two thresholds and keeps only cases whose outcome differs between the original and the altered rule; its near-miss kinds are tbd, which is why models that fail on other kinds do well on it.

Versions. rule_v1 was frozen, with hashes of every set, before the runs reported here, and is not modified. Rule-side items, author-written cases, rewritten cases and registered criteria are separate named sets, each frozen before any system was scored on it. Defects found after the freeze are listed with the affected items, and results are given with and without them (tbd items).

C Triplet Construction

Invariants. Every edit is checked on the target claim by executing the program: a flip must change the claim, a near-miss and a presentation edit must not, also for rules that combine criteria. Of the

proposed flips, 7.9% are discarded because a second criterion changes or the claim does not; of the proposed missing twins, 18.6% because the remaining inputs determine the claim. Near-misses are checked by executing the program on the state recovered from the rendered case, and the equalities $h_k(x) = 0$, $h_k(x^f) = h_k(x^n) = 1$ (or $h_k \equiv 1$ for measured inputs) are checked on the rendered text of every triplet. Each training corpus has 60,000 records; L2 has 2,000 triplets and the development set 300. No evaluation input contains the edit type, the role of the case, or the state.

Semantics. *Several mentions.* A criterion uses the mentions its predicate counts; for a measurement the most recent value dated at or before the reference date, for a finding any counted mention with status present. A counted “absent” and a counted “present” for the same finding and time make the state inconsistent, and such states are not generated. *Numbers.* Units are fixed per input and stated; where a unit conversion is required it is part of the rule text. Thresholds are inclusive or exclusive as the rule text says, and values exactly at a threshold form their own near-miss kind. *Time.* Every case has a reference date; “within W months” includes the boundary day. *Omission.* Measurements that are not given are unknown; history items that are not mentioned are absent, unless the case says they are unknown. This is a stipulation of the generator, not a general reading of clinical notes; tbd% of base cases state the required absence explicitly, and results on that subset are tbd. External datasets keep their own conventions.

Near-miss kinds. Numeric 40% (of which at the threshold tbd), other subject 20%, absent status 20%, time outside the predicate 20%. For criteria whose predicate counts past or family findings, such mentions are flips.

Harder cases, by requirement. We do not use an undefined “hard” label. Table 9 lists the requirements and how many test triplets have each.

Rule-side items. For a fixed case, two rules differ only in the applicability clause: time window, currency, subject, or boundary inclusivity. Each item has three cases (the contested form, an uncontested positive, an absent finding) and is solved only if all six judgments are right. A scorer that ignores the rule text, one that never counts the contested form and one that always counts it solve no item by construction (tbd items; tbd phrasings per clause).

Requirement	Example	<i>n</i>
Rule-dependent time scope	the same past event counts under “ever”, not under “within six months”	tbd
Competing mentions	a relative has the finding; the patient explicitly does not	tbd
Temporal selection	an older value crosses the threshold; the latest does not	tbd
Composition	two conditions with an explicit exception	tbd
Numerical semantics	values at inclusive and exclusive boundaries; unit given in the rule	tbd
Long note	15 or more unrelated lines	tbd

Table 9: Requirements that define the harder cases.

Text tiers. *Author-written cases:* tbd groups written by the authors from state specifications, each checked by a second author. *Rewritten cases:* notes rewritten by Gemma-3-27B are accepted if two extractors recover the intended state (8.7% rejected). Only accepted groups are scored (Ledger-RM: tbd). Rejected rewrites may no longer express the intended state, so their original label cannot be reused; they are kept as a data-quality audit set and are not an evaluation set.

D External Tiers

Releases. MedEinst as released by its authors; NLI4CT-P as distributed for SemEval-2024 Task 2; MedPRMBench and CareQA at pinned revisions listed in the repository.

Registered eligibility criteria. Criteria are taken as written from registered trials (tbd criteria from tbd trials). We keep single-condition criteria that can be decided from stated facts: laboratory thresholds, events within a time window, current conditions, pregnancy or breastfeeding, allergy, concurrent medication, age bounds. For each we release the trial identifier, the exact text, whether it is an inclusion or exclusion criterion, any normalisation, the program and its boundary and time tests, and the reason for every exclusion. An author who did not write the program checks it against the sentence; this is a check of logic, not of medicine. Trials and criterion texts that occur in the TrialGPT annotations are excluded. The set is used for testing only; training on such criteria would be a separate experiment. Rule-side variants of a criterion are labelled as synthetic interventions on the original text. Claims are “The patient meets this criterion” and its negation.

TrialGPT annotations. The released file has 1,015 patient–criterion annotations over 53 patient summaries and 103 trials, with expert fields (eligibility label, relevant sentences) and, separately, GPT-4 outputs. Protocol, fixed before any system was scored: inputs are the patient summary as released and the criterion with its type; no GPT-4 field is ever an input; targets are the expert labels; the dataset’s categories are kept, with “not enough information” predicted when neither claim clears the acceptance threshold carried over from the rule-tier development set, and “not applicable” items reported separately; the adapter and prompts are those selected on rule-tier development data, and a patient-disjoint development portion (tbd patients) is used for interface decisions and excluded from the reported test portion. The expert sentences evaluate evidence selection; they do not validate the subject, status or time fields of a ledger. If eligibility examples are later used for training, every compared method receives the same added data. We report accuracy, macro-F1, class-wise errors, and precision and recall of the reader’s quoted sentences against the expert sentences; intervals resample patients, and we inspect variation across trials. The patient summaries are synthetic, so this is an external expert-annotated evaluation, not validation on clinical records. Results: backbone tbd, verdict only with blocks tbd and with triplets tbd, matched prose tbd, Ledger-RM with blocks tbd and with triplets tbd (macro-F1).

MedEinst. Claims are the two diagnoses. For training rows, the found targets are the findings in which the two cases differ, taken from the structured evidence where released and from a token diff otherwise; the two agree on 96.2% of a sample of 500 pairs, and in 97 of 100 pairs inspected by the authors the difference names the edited finding. Other ledger fields are not supervised there. In the disease-held-out split a pair is in the test set only if both of its diagnoses are test diseases (12 of 49), and in the training set only if both are training diseases.

Key pairs. Pairs are mined within a split and reduced to a one-to-one matching, so that no question occurs in two pairs. Excluded: negatively phrased lead-ins, options that refer to other options, and questions without a case description. For training rows, ledgers are sampled once from the frozen rule-trained reader and kept if their quotations occur in the case and the frozen rule-trained judge

returns the keyed verdict on all four cells (38.5% kept); the kept set is fixed and shared by all ledger formats. By tercile of vignette similarity, reversal of the critic is 55.7%, 44.1% and 33.8%. CareQA covers six subject areas; results by area are in the repository, and the biology and chemistry areas contribute 4% of pairs.

CondMedQA. CondMedQA (Parekh et al., 2026) has 100 questions that contain a patient condition, each with the answer given the condition and the general answer that would hold without it; candidates were generated by a model, verified by its authors, and 30 were reviewed by annotators with medical expertise. We score both answers as claims for the question as posed and count how often the condition-specific answer is preferred: 41.0% for the critic, 56.0% for rule-only Ledger-RM [46, 66] and 80.0% for the closed judge. This is a test of resisting the default answer and not a reversal pair, since the benchmark has no version of the question without the condition; no rule is stated, so it also measures knowledge.

Claim-side applicability on real notes. EHRNote-ChatQA (Kim et al., 2026) asks questions about a patient’s discharge summaries, each with one correct and four incorrect answers that follow defined patterns and were reviewed by medical experts. Two patterns mirror our near-misses on the claim side: a correct fact attributed to the wrong admission, and a related entity or attribute taken from elsewhere in the notes. Scoring each answer as a claim against the notes, the correct answer is preferred to such a distractor in 58.0% of cases by the critic, in 69.5% by rule-only Ledger-RM and in 77.0% with clinical pairs. The data are under credentialed access and may not be sent to external services, so closed judges are not evaluated on them.

NLI4CT-P. Base macro-F1 is 71.0 for the critic and 76.4 for rule-only Ledger-RM; joint correctness over an original statement and all its interventions is 38.5% and 47.9%. Faithfulness and consistency follow the task definitions; pairs that share an original statement are resampled together.

Overlap. By normalised question text, CareQA shares no question with MedQA. The MedQA portion of ReMedQA (Cocchieri et al., 2026) lies within the MedQA test split; we do not use ReMedQA.

System	Key pairs	NLI-F	NLI-C
<i>Training-free, Qwen3.5-9B</i>			
Critic	44.5	60.2	83.0
+ default correction	49.3	64.9	80.2
+ prompted ledger	48.0	66.1	82.4
Generated-program verifier	—	—	—
<i>Trained without medical QA data (clinical columns are zero-shot)</i>			
Verdict only, FoVer data	43.8	63.5	84.0
Verdict only, rule triplets	49.0	65.2	84.1
Evidence summary, rule triplets	51.6	68.0	85.2
GenPRM-style verifier, rule triplets	50.2	66.9	84.8
Ledger-RM, rule balanced	50.1	67.4	82.6
Ledger-RM, rule blocks	53.5	71.2	79.4
Ledger-RM, rule triplets	52.9	69.8	86.0
<i>With medical data</i>			
Ledger-RM, rule triplets + step-error data	55.3	71.5	86.1
Ledger-RM, clinical pairs only	72.9	80.7	87.5
Ledger-RM, rule triplets + clinical pairs	75.6	83.9	88.2
<i>References</i>			
Claude Opus 5.5, zero-shot	82.4	87.2	90.3
Claude Opus 5.5, prompted ledger	84.1	89.5	90.8
Extraction + hand-written program	—	—	—

Table 10: Rows of Table 4 on key pairs (reversal) and NLI4CT-P (faithfulness, consistency), %.

E Ledger Format and Scoring

```

need: penicillin allergy in the patient
found: "penicillin allergy (anaphylaxis)"
subject: other (mother)
status: present
time: current

```

Given this ledger, the rule and the claim “Order amoxicillin”, the judge returns +. Malformed ledgers occur in at most 0.17% of rule-tier and 2.4% of MedEinst instances; a malformed ledger is recorded as an invalid outcome and both claims are rejected. Neither record format may state a verdict: on generated records a decision word or the text of a claim occurs in **tbd%** of prose records and **tbd%** of ledgers. Temperature scaling for the combined reward is fitted on development data separately for the two scores. The verification pass prompts the same adapter with one ledger entry and the case and asks whether value, subject, status and time are as recorded; it rejects 41% of wrong entries and 3.2% of correct ones. In triplet and pair evaluations one ledger serves both claims. In selection the reader receives a step; for two candidate steps about the same criterion the ledgers are identical in 98.7% of cases. On L2 the reader is right on 96.9%, 95.4% and 94.1% of found, subject and time fields. In 2.1% of L2 cases it quotes a span that occurs in the case and is not the decisive finding; the string check cannot detect this.

F Diagnostics

Near-miss decisions. Hold is near-miss correctness. To keep wrong-but-consistent decisions from

Scorer (L2)	Rev	Hold	TA
Always default	0	100	0
Claim only	0	–	0
Concept named (h_k)	≤ 45	–	0
Bag of words, case + claim	tbd	tbd	7.1
Attribute-blind, logistic	tbd	tbd	40.0
Trigger lexicon + program, training cue phrases	tbd	tbd	54.0
given the test cue phrases	tbd	tbd	99.5
General-purpose trigger tagger + program	tbd	tbd	tbd

Table 11: Shortcut scorers on L2 (%); triplet accuracy measured. Entries of the first three rows follow from Proposition 1. Concept named: any deterministic scorer that uses the case only through h_k . Attribute-blind: a logistic model on concept–value features without subject, status and time; it solves numeric triplets only. Trigger lexicon: a rule-based reader with negation, experimenter and temporality triggers in the style of ConText (Harkema et al., 2009), followed by the rule program. Built from training cue phrases it reaches 54.0%; given the test cue phrases it reaches 99.5%, which is a reference and not a baseline. The last row is a frozen general-purpose implementation, which knows more synonyms than our training phrases. Trained models that exceed these scorers outperform the shortcut implementations tested here; this does not show that they use no shortcut.

Signal	Base	Ties	UR (n)	P	K	G
Inj.-error PRM	93.5	0.0	8.1 (4,769)	92.8	89.5	71.4
Qwen3.5-9B	80.2	0.0	24.6 (4,090)	95.3	91.2	56.8
Qwen3.5-27B	88.4	0.0	26.0 (4,508)	97.6	95.4	33.4
Claude Opus 5.5	96.0	3.8	12.9 (4,896)	99.2	98.1	11.3

Table 12: L0 (%). Base accuracy; tied preferences; unnecessary reversal among triplets with a correct base preference, with its denominator; reading and application reversal; composition gap.

counting as desirable behaviour we also report, per system, the share of base and near-miss pairs with the same decision, the share with both decisions correct, and near-miss correctness given a correct base (Table 13, with denominators per kind).

System	thr.	neg.	num.	subj.	time
Verdict only, blocks	33.8	93.2	92.0	11.5	60.5
Ledger-RM, blocks	72.0	97.5	99.8	48.5	58.8
Verdict only, triplets	93.2	91.2	92.8	91.0	88.2
Ledger-RM, triplets	97.8	or above for every kind			
n (triplets)	tbd	tbd	tbd	tbd	tbd
Same decision on base and near-miss			tbd		
Both correct			tbd		
Near-miss correct given base correct			tbd		

Table 13: Triplet accuracy on L2 by near-miss kind (%; seed 0): value at the threshold, negated finding, value near the threshold, other person, past finding. Lower rows: quantities to be reported per system.

Shift relative to a missing-input baseline. Let $d_0 = d(x^m)$, $\delta(x) = d(x) - d_0$, and $y \in \{+1, -1\}$ the correct sign of $d(x)$. For cases with $y d_0 < 0$, where the baseline preference opposes the label, $y d(x) = -|d_0| + y \delta(x)$, so the preference is right if and only if

$$y \delta(x) > |d_0|. \quad (3)$$

Cases with $d_0 = 0$ have no opposing baseline and are counted separately. We sort the others into four disjoint classes, in this order: *crossed* (Eq. 3); *unmoved* ($|\delta| \leq \tau$, the model’s median $|d(x^p) - d(x)|$ over presentation edits); *wrong direction* ($y \delta < 0$); *short*. The split restates the decision rule and adds the direction and relative size of the shift. The *shift ratio* compares that size with the model’s response to an error inside a step:

$$\kappa = \frac{\text{med}\{y \delta(x) : y d_0 < 0\}}{\text{med}\{u(s^+, x) - u(s^-, x)\}}, \quad (4)$$

where the denominator runs over matched pairs of a correct step s^+ and its injected-error version s^- on the same case in the MedPRMBench development split; both medians are in log-odds, and κ is reported with a bootstrap interval when the denominator is positive. For the injected-error PRM, 16.8% of baseline-conflicting cases cross, 55.2% move the right way and fall short, 19.5% do not move and 8.5% move the wrong way; $\kappa = 0.18$ (Figure 4).

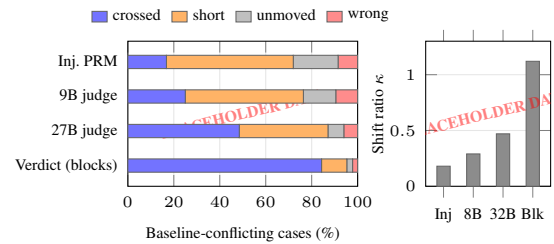


Figure 4: Rule tier. Left: cases whose correct preference opposes the missing-input baseline, by class (Eq. 3; models that expose log-probabilities). Right: shift ratio κ (Eq. 4). Verdict (blocks): the verdict-only model trained on flip blocks.

Readouts and sampling noise. Dedicated PRMs use their released scoring head, open judges the log-odds of the verdict token, and API judges without log-probabilities the verdict frequency over 32 samples at temperature 1.0, for which the standard error of a frequency is at most 0.09. Scoring

Qwen3.5-27B by the same sampling protocol lowers its triplet accuracy from 45.3% to 42.0%, with 4.1% ties.

Input ablations. Replacing the question stem by “No case description is available” lowers PRM-Score on MedPRMBench by 1.2–2.9 points for trained PRMs. This is an input ablation and not a measure of grounding: many injected errors are detectable within the step, and chains restate patient facts. On the error types that the benchmark defines relative to the case the loss is 4.0–9.5 points, and 6.2–13.0 when numbers and quoted spans of the stem are also masked in the chain.

Exploratory. A probe on the residual stream of the critic predicts the criterion outcome on held-out signature classes with 90.8% accuracy (control task (Hewitt and Liang, 2019): 49.6%); decodability does not show use (Elazar et al., 2021).

G Additional Results

Training. LoRA rank 64, batch size 64, one epoch over 60,000 records, learning rate $1e-4$, $\lambda = 1.0$, $p = 0.3$ [confirm against the run manifest that ledger resampling was used]. Rows with clinical pairs mix rule triplets and pairs 1:1. A verdict-only run takes about 2.2 GPU-hours and a two-stage run about 2.8, including evaluation; peak GPU memory in the configuration with on-GPU checkpointing was 21.9 GB, which we report for that configuration only.

Resources. Per scored claim: verdict only, 1 pass and 1 generated token; Ledger-RM, 2 passes and 45 tokens; generated-program verifier, 1 pass, 540 tokens and one execution; GenPRM-style verifier, 1 pass, 500 tokens and one execution; closed judge with ledger prompt, 1 API call and 210 tokens, about 60 times the list price of the backbone per claim. Latency and hardware are in the repository.

Policy training. This is the only experiment in which a policy is optimised against a learned reward, and the only basis for statements about reward exploitation. GRPO from one Qwen3.5-9B initialisation on rule triplets, 3 runs, 8 rollouts per prompt, 2000 steps, advantages standardised within a group. Before each run we check that reward terms are not collinear within groups and that the setup fits a 64-prompt subset. Accuracy on base-flip pairs of held-out structures, judged by the rule program, moves from 38.6% to 52.3% with an out-

Variant (rule triplets only)	Rev	Hold	TA	MR	ME
Ledger-RM (facts only)	99.2	100.0	99.2	≈100	44.8
+ decision field	tbd	tbd	tbd	tbd	44.1
decision field only	tbd	tbd	tbd	tbd	38.2
program on predicted bit	tbd	tbd	tbd	tbd	–
reader writes bit only	tbd	tbd	tbd	tbd	34.5
condition derived by the reader	tbd	tbd	tbd	tbd	tbd
+ verification pass	tbd	tbd	tbd	tbd	45.4
program-supplied ledger	tbd	tbd	tbd	tbd	–
program on predicted ledger	tbd	tbd	tbd	tbd	–
One-stage ledger	tbd	99.6	91.0	≈100	47.5
Value ledger	tbd	tbd	tbd	tbd	42.0
Evidence summary (matched prose)	tbd	tbd	98.0	≈100	39.4
judge also sees the case	tbd	tbd	tbd	tbd	tbd
Concept scorer	tbd	tbd	tbd	tbd	31.0
Premise gate	tbd	tbd	tbd	tbd	30.5
– near-misses	98.7	76.2	75.3	≈100	46.3
+ probe re-weighting	tbd	tbd	tbd	tbd	44.0
– other-person near-misses	tbd	tbd	tbd	tbd	tbd
– past near-misses	tbd	tbd	tbd	tbd	tbd
– threshold near-misses	tbd	tbd	tbd	tbd	tbd
– presentation edits	tbd	tbd	tbd	tbd	44.5
– ledger resampling	tbd	tbd	tbd	tbd	44.0
Verdict only, pairwise loss	tbd	tbd	tbd	tbd	33.8

Table 14: Ablations on L2 and MedEinst zero-shot (%); black: measured, seed 0. Decision field: the reader also writes whether the condition applies. Condition derived: the reader is given the rule and the claim and must find the condition under test. Verification pass: every entry is re-read against the case and regenerated once if rejected, after Agand (2026). Program-supplied ledger: the judge receives the target ledger; reported with paired transitions in the text. Program on predicted ledger: the rule program replaces the judge; it uses a program at test time. Concept scorer: fields predicted by linear heads and a linear verdict. Premise gate: the step check, attenuated toward a neutral score when the finding a step relies on is not supported by a ledger extracted once per case, after Wang et al. (2026). Probe re-weighting: near-misses are not trained on; base-flip pairs whose preference shifts under near-miss and presentation edits are down-weighted; this is our adaptation of Liu et al. (2026) to case edits, not a reproduction. Rows that remove one near-miss kind are evaluated on triplets of that kind. Pairwise loss: Bradley–Terry on the two claims.

come reward, 33.9% with the injected-error PRM and 63.5% with Ledger-RM. A reference-graph reward after Mu et al. (2025), which scores a rollout by its coverage of the decisive findings and of the chain from finding to criterion to conclusion, gives 58.0%; the rule program supplies the graph, so this reward, like the outcome reward, uses the answer for each case.

Selection protocol. A trace is eligible if it contains a final answer (6.1% are not). Ranking by the product of the two calibrated scores gives 78.0% on MedQA and 66.1% on key pairs, a logistic combination fitted on development data 78.5% and

System	Dev	L0	L1	L2	L3-inv	L3-alt	Long
Critic	tbd	tbd	tbd	tbd	tbd	tbd	tbd
Verdict only, blocks	67.3	68.8	73.8	58.2	62.9	94.6	59.1
Ledger-RM, rule blocks	70.3	73.5	78.2	75.3	70.1	85.3	76.4
Verdict only, triplets	99.7	99.0	97.6	91.3	96.9	100.0	90.5
Ledger-RM, rule triplets	tbd	tbd	tbd	99.2	97.9	100.0	98.3
Closed judge, ledger	tbd	tbd	tbd	tbd	tbd	tbd	tbd

Table 15: Rule ladder (triplet accuracy, %); black: measured, seed 0. Dev: trained rules, test wording; L3-inv: invented rules; Long: longer notes with superseded values (requirements in Table 9). On reading and application pairs the first three trained systems reach 77.7%, 76.0% and 99.4%. Macro-averages over rules and over signature classes, per-seed values and the lowest held-out class are tbd.

Comparison	Isolates
Verdict vs. one-stage ledger	intermediate supervision
One-stage vs. two-stage ledger	separating from judgment
Two-stage ledger vs. matched prose	typed structure
Case-blind vs. case-visible judge	restricting access to the case
Own vs. program-supplied ledger	what better extraction could recover
Learned judge vs. rule program on the same ledger	what the neural judge adds

Table 16: Comparisons among interfaces and what each isolates.

Configuration	Cases	Reader	Judge	Tokens	Steps
Verdict only	tbd	–	tbd	tbd	tbd
Rationale	tbd	–	tbd	tbd	tbd
Evidence summary	tbd	tbd	tbd	tbd	tbd
Applicability ledger	tbd	tbd	tbd	tbd	tbd

Table 17: Supervision and compute per configuration: unique cases, reader targets, judge targets, training tokens, optimiser steps. Equal record counts do not by themselves mean equal supervision.

H Development History and Error Analysis

The first generator and the shortcut it allowed.

[Authors: write this from your own records.] The pilot generator had 11 rules and three templates per mention form, shared between training and development data. A verdict-only model trained on its triplets solved every development triplet. On rules held out from training in which a past or a family finding counts, it solved 59.7%: with past and family mentions never counting in training, “past or family means ignore” was a sufficient strategy. This result is from the pilot setup and motivates the rule-side test; it is not a failure shown for the current library.

What changed. rule_v1 holds out signature classes, templates and cue phrases; contains rules under which past or family findings count; adds presentation edits and missing inputs to every corpus; and was frozen before the runs reported here. Rule-side items test the same failure directly and were built after the freeze as a separate set.

Errors of the trained systems. From tbd sampled failures, stratified by system and near-miss kind and coded by the authors: extraction errors tbd, applicability errors tbd, wording not seen in training tbd, rule misread tbd, other tbd.

67.0%, and the minimum 78.3% and 66.7%. With a paraphrased vignette the combined reward gives 78.0% on MedQA. For nested pools the selected reward cannot fall as N grows, so we report selection regret, the gap between oracle and selected accuracy: at $N = 64$ it is 27.0 points for the step check and 15.5 for the combined reward on key pairs. On the open-ended version of CareQA, graded against reference answers by a fixed model grader, the share graded correct moves from 48.2% to 51.0%; this check is exploratory.